

Use Case: “Pay Order”

Use case code	UC00X	Use case name	Pay Order
Actors	Customer, VNPay		
Description	Allow customer to pay for the order		
Preconditions			
Basic Flow of Events			
	No.	Performed by	Action
	1.	AIMS	Display the invoice (Table 5)
	2.	Customer	Confirm and accept to pay the invoice
	3.	AIMS	Redirect to VNPay
	4.	VNPay	Send the result back to AIMS
	5.	AIMS	Save invoice and payment transaction
	6.	VNPay	Notify the payment result
Alternative flows			
	No.	Performed by	Action
	5a.	Customer	Customer cancels the payment transaction
Postconditions	None		

1. Use Case Code: UC002

2. Brief Description:

This use case describes the interaction between the AIMS software with the customer and VNPay when the customer chooses to pay for an order. The customer will be presented with an invoice and then redirected to VNPay for the payment process.

3. Actors:

3.1. **Customer:** The person who interacts with the AIMS software to make the payment for their order.

3.2. **VNPay:** The payment gateway that processes the customer’s payment.

4. Preconditions:

- The order is pending payment.

5. Basic Flow of Events:

1. The **AIMS software** displays the invoice to the customer (see Table 3).
2. The **customer** selects the option to pay the invoice.
3. The **AIMS software** redirects the customer to the VNPay payment gateway.
4. **VNPay** processes the payment and returns the payment result (success/failure) to AIMS.
5. The **AIMS software** saves the invoice and payment transaction details.
6. The **AIMS software** notifies the customer of the payment result (success or failure).

6. Alternative Flows:

Table 1 - Alternative flow of events for UC “Pay Order”

No	Location	Condition	Action	Resume Location
1.	At Step 5	The customer cancels the payment transaction	The software returns to Step 1 (Invoice display)	Step 1
2.	At Step 4	Payment fails	The software notifies the customer and returns to Step 2 (Payment selection)	Step 2

7. Input Data:

Table 2 - Input Data for Payment Information

No	Data Fields	Description	Mandatory	Valid Condition	Example
1.	Customer's order details	Items, total price, shipping fees, etc.	Yes	Must be accurate	Book x2, 246,000 VND
2.	Payment method	Payment processed via VNPay	Yes	Must be VNPay	VNPay

8. Output Data:

Table 3 - Output Data of Invoice

No	Data Fields	Description	Display format	Example
1.	Title	Title of a media product	Text	Harry Potter movie disc
2.	Price	Price of the corresponding media product	Comma for thousands separator, Positive integer, Right alignment	100,000
3.	Quantity	Quantity of the corresponding media	Positive integer, Right alignment	2
4.	Amount	Total amount for the corresponding media product	Comma for thousands separator, Positive integer, Right alignment	200,000
5.	Subtotal Before VAT	Total price of all products in the cart before VAT	Comma for thousands separator, Positive integer, Right alignment	2,000,000
6.	Subtotal After VAT	Total price of all products in the cart with VAT	Comma for thousands separator, Positive integer, Right alignment	2,000,600
7.	Shipping Fees	The cost of shipping based on delivery method and address	Comma for thousands separator, Positive integer, Right alignment	30,000
8.	Total	Sum of the subtotal (with VAT) and shipping fees	Comma for thousands separator, Positive integer, Right alignment	2,000,600
9.	Currency	The currency for the total amount	Text	VND
10.	Customer Name	The name of the customer	Text	Nguyen Van A
11.	Phone Number	The customer's phone number	Phone number format	0999999999
12.	Province	The province/city where the shipping address is located	Text	Hanoi
13.	Address	The delivery address	Text	12, 34 Hoan Kiem

14.	Shipping Instructions	Any additional instructions for the delivery	Text	Deliver between 9 AM - 5 PM
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9. Postconditions

- The payment is successfully processed, and the order status is updated.
- The payment result (success/failure) is recorded.